

being received, the decoder evaluates the time value that it expects (based on the decoder's timer and from the encoder's last-saved time-stamp) with the time-stamp from the encoder. If the two values match, the decoder takes appropriate action (that is, opens a door).

However, the receiver/decoder side needs to support not just one but many more such encoders. So it needs to have different time values for all encoders paired with the decoder. Obviously, the decoder cannot implement many such timers.

The solution is to use a single timer on the decoder side and use time delta values for each encoder. These time delta values for each encoder need to be saved in non-volatile memory. When a data packet arrives from the encoder, the decoder would read the saved time delta value for that particular encoder and use its current timer value to estimate the timer value sent by the encoder. The decoder would, again, compare the two time values and if these match (or there are acceptable small differences), it would take appropriate action.

Synchronisation

The whole synchronisation mechanism is built around free-running timers, both on the encoder and the decoder sides. Ideally, these timers run synchronously and are in constant synchronisation. However, all time-keeping devices have limited stability and precision. Devices that feature high stability and precision can be very expensive. So choosing the right time-keeping mechanism is more a design trade-off. A typical implementation will use the same time-keeping circuit on both encoder and decoder sides. A cost-effective solution can be using a crystal oscillator. However, even if the same crystal is used at both ends, there will be

some frequency differences.

Discrepancies in frequency.

First, there is the initial crystal precision. This is specified by the crystal manufacturer and it is typically in the range of 5ppm to 40ppm (parts per million). The initial crystal precision is due to the manufacturing process. The crystal frequency cannot exactly be the specified value, but a very close one. This is specific and constant over time for every crystal.



MCS3142 high-security encoder from Microchip Technology

Second, crystals are sensitive to temperature. The oscillating frequency is specified at 25°C. Between -20°C and 50°C, frequency can vary up to an additional 20ppm.

Third, we must also take into consideration crystal aging. This is less important, since it happens over a very large period of time. Taking into account all the above tolerances will mean that timers on both the encoder and the decoder will never be in perfect synchronisation, except at first pairing. The decoder will not be able to calculate an exact match of the encoder's real-time value. It will only be able to calculate an approximate value.

The decoder can still be synchronised with all encoders by using an acceptance window. For example, once synchronised, timer error between the encoder and the decoder will be at minimum. As time passes, time drift between the encoder and the decoder will increase. It is then up to the decoder to decide the maximum time difference for a packet to be accepted as a valid packet.

Re-synchronisation

Since the system relies on the fact that the timers on both ends are in constant synchronisation, it is very important that the timer's operation is not interrupted. There are cases when synchronisation can be lost due to power failure (like a mains grid incident or battery replacement).

On the encoder side, this will typically happen when the battery gets exhausted and needs replacement. This can be easily resolved by manually re-synchronising the encoder. On the decoder side, a loss of power is more serious, especially since the decoder needs to keep track of multiple encoders.

Re-learning all encoders is feasible only when there is a relatively low number of encoders. If the system has more than a few encoders, manually synchronising all encoders is not a good option. In such cases, it is recommended that the decoder has a time-keeping back-up circuit such as a real-time clock circuit with a battery backup.

Conclusion

Proper implementation can offer a very high security level. System attacks like capture-and-reply and radio frequency jamming become basically useless.

Another important advantage is that such an encoder can be used with multiple decoders. This is because the decoder only verifies time synchronisation and the authenticity of the encoder. Overall, this is a high-security and cost-effective security solution.

From the user standpoint, the solution introduces no difference and allows for ease of use. This technology can easily be used in conjunction with an authentication scheme so it can also provide a good level of OEM protection. **EFY**

The technique described in this paper is patented by Microchip Technology Inc.